VOLUME 12 — NID DAT LAST-MINUTE REVISION BOOK

Contents: 1. One-page exam strategy · 2. Visual & design principles · 3. Colour · 4. Typography & visual communication · 5. Perspective & drawing rules · 6. Reasoning formulas · 7. Indian art, crafts & textiles · 8. Architecture · 9. Designers & movements · 10. Logos & products · 11. Terms · 12. **Flashcards (160)** · 13. **Top 100 NID DAT mistakes**

1. ONE-PAGE EXAM STRATEGY (read this the night before)

- **Exam:** Sunday 20 Dec 2026 · 3 hrs · 100 marks · pen and paper · no negative marking · Prelims = **shortlisting only**.
- **Two possible formats:** (A) 22 objective + 4 subjective (the Handbook scheme); (B) all-subjective, about 6 questions with sub-parts (reported for 2026). Adapt within the first 5 minutes.
- **First 5–10 minutes:** read everything; **circle the counts, viewpoints, users and colour limits**; write a time budget next to each question; jot 2 idea keywords per question.
- **Time rule:** minutes ≈ marks × 1.65. Use the **70% rule**: move on when a drawing is 70% done.
- **Never leave a question blank.** A 3-minute labelled sketch scores marks; a blank scores zero.
- **The order of value:** idea → clarity → composition → proportion → rendering.
- **MCQs:** answer all of them (no negative marking). Skip anything over 90 seconds, then come back.
- **Last 15 minutes:** fill unanswered MCQs, add labels and titles, add colour to the strongest answer, check the counts.

2. VISUAL & DESIGN PRINCIPLES (core list)

- **Elements:** point, line, shape (2D), form (3D), colour, texture (actual or visual), space (positive or negative), value.
- **Principles:** balance (symmetrical / asymmetrical / radial), contrast, emphasis, hierarchy, proportion (part to part), scale (relative to the human or another object), rhythm (regular / alternating / progressive / flowing), repetition, movement (the eye path), unity, variety, harmony.
- **Gestalt:** proximity, similarity, closure, continuity, figure–ground, common fate, symmetry, prägnanz.
- **Composition:**
  - the rule of thirds
  - one focal point
  - odd numbers of subjects
  - lead room in front of a moving subject
  - leading lines
  - framing
  - foreground, middle ground and background
  - avoid tangents
  - don't put the horizon in the middle
- **Design thinking (Stanford):** Empathise → Define → Ideate → Prototype → Test. **Double Diamond:** Discover → Define → Develop → Deliver.
- **Don Norman:** affordance, signifier, feedback, mapping, constraint. **Dieter Rams:** "Less, but better", 10 principles.
- **Universal design:** works for the widest range of users (the curb-cut effect). **Ergonomics:** design for the 5th–95th percentile.

- **Sustainability:** refuse, reduce, reuse, repair, recycle, rot; circular economy; cradle to cradle; design for disassembly; mono-material packaging; avoid greenwashing.

3. COLOUR (essentials)

- **RYB primaries:** red, yellow, blue → secondaries orange, green, violet → tertiaries such as red-orange.
- **RGB (light, additive):** R+G = yellow, G+B = cyan, R+B = magenta, all three = white.
- **CMYK (print, subtractive):** C+M = blue, M+Y = red, C+Y = green.
- **Complements (RYB):** red–green, blue–orange, yellow–violet.
- **Tint** = + white; **shade** = + black; **tone** = + grey.
- **Hue** = the colour family; **saturation** = intensity; **value** = lightness.
- **Schemes:** monochromatic, analogous, complementary, split-complementary, triadic, tetradic.
- **60-30-10 rule.** Warm colours advance and cool colours recede. **Atmospheric perspective:** distant objects look paler and bluer.
- **Accessibility:** about 8% of men are red–green colour-blind. Use value contrast plus shape, never colour alone.
- **Safety colours:** red = prohibition, yellow = warning, blue = mandatory, green = safe or exit.
- **India:** saffron = sacred and courage; white = mourning (in many communities); red = bridal and auspicious.

4. TYPOGRAPHY & VISUAL COMMUNICATION

- **Serif** (Times, Garamond, Bodoni, Baskerville) · **Sans** (Helvetica, Futura, Gill Sans, Arial, Univers, Frutiger) · **Slab** (Rockwell) · Script · Display · Monospace.
- **Kerning** = one letter pair · **Tracking** = a whole word or line · **Leading** = line spacing · **Counter** = the enclosed space in a letter · **x-height** · baseline · ascender and descender.
- **Legibility** = telling characters apart (a property of the typeface); **readability** = reading blocks of text easily (a property of the layout).
- **Rules:** at most 2 typefaces; 45–75 characters per line; use sans serif with high contrast for signage.
- **Pictograms:** a simple silhouette, a consistent stroke and grid, one idea per sign, no text needed, a red circle-and-slash for prohibition.
- **Poster:** headline (3–6 words) → one strong image → supporting line → call to action → logo, date and venue.
- **Logo types:** wordmark, lettermark, pictorial, abstract, mascot, emblem, combination. A good logo is simple, memorable, scalable, works in mono, and is appropriate.
- **Packaging functions:** protect, preserve, contain, communicate, convenience, sustainability.

5. PERSPECTIVE & DRAWING RULES

- **HL** = eye level. **1-point** (frontal: roads, rooms) · **2-point** (corners; verticals stay vertical) · **3-point** (looking up or down).
- **Standing viewer:** the heads of all standing adults sit on the HL, at any distance. A child half an adult's height has its head **halfway between its feet and the HL**.
- **Bird's-eye** = a high horizon with a third VP below; **worm's-eye** = a low horizon with a third VP above.
- **Human proportions:** an adult is 7.5–8 heads tall; the midpoint is the hips; the fingertips reach mid-thigh; the eyes sit at half the head's height.
- **Size references:**

Object	Approximate size
Door	2.1 m
Adult	1.6–1.75 m
Chair seat	0.45 m
Table	0.75 m
Kitchen counter	0.85 m
Bus	3 m tall
Auto-rickshaw	1.7 m tall

- **Shadows:** fall away from the light; are long in the morning and evening and shortest at noon; morning shadows point west.
- **Materials:** glass = sharp highlights and see-through; steel = high contrast with reflections; wood = grain; fabric = folds from tension points.
- **Storyboards:** plan the beats; keep characters and setting consistent; vary wide, medium and close shots; follow the 180° rule; number the frames; make the last frame show the resolution or a design improvement.
- **Product sketch:** a 3/4 view + 1–2 detail views + labels + a hand for scale + a small in-use scene + a 2-line justification.

6. REASONING FORMULAS & RULES

- **Mirror clock:** 11:60 − time (use 23:60 − time for times from 11:00 to 12:59).
- **Vertically symmetric letters:** A H I M O T U V W X Y. **Horizontally symmetric letters:** B C D E H I K O X. **Both:** H I O X.
- **Letters that look the same after a 180° rotation:** H I N O S X Z.
- **Folding:** n folds → 2^n layers and holes. Unfold in **reverse** order. A fold along y = x swaps the coordinates; a fold along x + y = s gives (s − y, s − x).
- **Painted cube (n):** 8 corners (3 faces); 12(n−2) edges (2 faces); $6(n-2)^2$ face-centres (1 face); $(n-2)^3$ inside (0 faces).
- **Dice:** opposite faces sum to 7. **Nets:** in a row, faces one apart are opposite; in a 1-4-1 net, the square above and the square below are opposite. A cube has **11** nets.
- **Euler:** V − E + F = 2.
- **Counting:**
  - squares in an n×n grid = Σk^2; in a 4×4 grid = 30; in a 5×5 grid = 55
  - rectangles in an m×n grid = C(m+1, 2) × C(n+1, 2)
  - triangles from a vertex with k lines and b bases = C(k, 2) × b
  - a square with both diagonals = 8 triangles; adding both midlines gives 16
- **Gears:** meshing gears alternate direction; teeth × turns is constant; an odd closed loop jams. An open belt keeps the direction; a crossed belt reverses it.
- **Clock angle:** $|30H - 5.5M|$. Handshakes: n(n−1)/2. Diagonals of a polygon: n(n−3)/2.
- **Interior angle sum:** (n−2) × 180°. Only triangles, squares and hexagons tessellate alone.
- **Scaling:** lengths × k → areas × k^2 → volumes × k^3.
- **A-paper:** A0 = 1 m^2; the sides are in the ratio 1 : √2; 1 A1 = 8 A4. **Golden ratio** ≈ 1.618.
- **Percentages:** successive changes a and b give a + b + ab/100; +20% then −20% = −4%. Profit % is on cost.
- **Speed:** km/h × 5/18 = m/s. Average speed over equal distances = 2ab/(a + b).

7. INDIAN ART, CRAFTS & TEXTILES (rapid list)

- **Paintings:**
  - Madhubani (Bihar: double lines, filled space)
  - Warli (Maharashtra: white on mud; circle, triangle, square)
  - Gond (MP: dots and dashes; Jangarh Singh Shyam)
  - Pithora (Gujarat/MP)
  - Bhil (dots; Bhuri Bai)
  - Saura (Odisha)
  - Pattachitra (Odisha: Raghurajpur)
  - Kalighat (Kolkata: bold lines, satire)
  - Kalamkari (AP: Srikalahasti pen / Machilipatnam block)
  - Phad (Rajasthan: Pabuji scrolls, Bhopas)
  - Tanjore (TN: gold foil)
  - Cheriyal (Telangana)
  - Manjusha (Bihar: snakes)
  - Sohrai-Khovar (Jharkhand)
  - Thangka (Buddhist)
- **Miniatures:** Mughal (Akbar's *Hamzanama*, Mansur's birds), Rajput (Kishangarh "Bani Thani" by Nihal Chand), Pahari (Kangra, Basohli).
- **Modern art:**
  - Raja Ravi Varma (oleographs)
  - Abanindranath (*Bharat Mata*)
  - Nandalal Bose (the Constitution; the Haripura posters)
  - Amrita Sher-Gil
  - Jamini Roy
  - the Progressive Artists' Group (Bombay, 1947: Souza, Raza with his "Bindu", Husain)
  - Ram Kinkar Baij (*Santhal Family*)
- **Saris and weaves:**
  - Banarasi (UP)
  - Kanjeevaram (TN, korvai borders)
  - Paithani (Maharashtra, peacocks)
  - **Patola** (Patan, double ikat)
  - Pochampally (Telangana, ikat)
  - Chanderi and Maheshwari (MP)
  - Baluchari (WB, narrative pallu)
  - Jamdani (supplementary weft)
  - Muga (Assam, golden)
  - Eri ("ahimsa" silk)
  - Kota Doria (checks)
  - Ilkal (Karnataka)
  - Kasavu (Kerala)
- **Embroidery:** Chikankari (Lucknow, white on white), Phulkari (Punjab), Kantha (Bengal, running stitch, upcycling), Kasuti (Karnataka), Zardozi (metallic), Toda (Nilgiris), Pipli appliqué (Odisha), Lambani (Banjara).
- **Dyeing and printing:** Bandhani (dots), Leheriya (waves), Ajrakh (indigo + madder), Bagru and Dabu (mud resist), Sanganeri, Bagh, Batik (wax). **Ikat dyes the yarn before weaving.**
- **Crafts:**
  - Blue Pottery (Jaipur, quartz)
  - Bidriware (Bidar, zinc alloy + silver)
  - Dhokra (lost wax)

    - Channapatna (lacquered wood toys)
    - Kondapalli
    - Aranmula Kannadi (a metal mirror)
    - Pietra dura (Agra)
    - Bankura horse
    - Longpi (Manipur, no wheel)
    - Thewa
    - Meenakari
    - Rogan (Nirona, Kutch)
    - Sankheda furniture
    - Sikki grass
    - Kathputli
- **GI:** Act of 1999, in force from 2003; the first GI was **Darjeeling Tea (2004)**; the GI Registry is in **Chennai**.

8. ARCHITECTURE (rapid list)

- **Temple styles: Nagara** (shikhara: Khajuraho, Konark, Lingaraja), **Dravida** (vimana and gopuram: Thanjavur, Madurai, Mahabalipuram), **Vesara** (Hoysala: Belur, Halebidu).
- **Key works:**
    - **Kailasa, Ellora** (monolithic, Rashtrakuta Krishna I)
    - **Brihadeeswara** (Rajaraja Chola I, about 1010)
    - **Konark** (Narasimhadeva I; a chariot with 24 wheels)
    - **Rani ki Vav** (shown on the ₹100 note)
    - **Humayun's Tomb** (the first garden tomb)
    - **Taj Mahal** (Ustad Ahmad Lahauri, 1632–53)
    - **Gol Gumbaz** (a whispering gallery)
    - **Charminar** (1591)
    - **Hawa Mahal** (1799)
    - **Jantar Mantar**
- **Colonial:** CSMT (F.W. Stevens), Gateway of India (Wittet), Victoria Memorial (Emerson), Rashtrapati Bhavan and India Gate (Lutyens), North and South Blocks (Baker).
- **Modern:**
    - Chandigarh (Le Corbusier)
    - IIM Ahmedabad (Louis Kahn)
    - **B.V. Doshi**: Sangath, IIM Bangalore, Aranya; Pritzker 2018
    - **Charles Correa**: Gandhi Smarak Sangrahalaya, Jawahar Kala Kendra (navagraha plan), Kanchanjunga Apartments
    - **Laurie Baker**: low-cost brick, the rat-trap bond
    - the Lotus Temple (Fariborz Sahba)
    - the New Parliament (Bimal Patel)
    - the Statue of Unity (Ram Sutar, 182 m)
    - the NMIA (ZHA, the lotus, 2025)
- **World:**
    - Fallingwater and the Guggenheim NY (Wright)
    - Villa Savoye, the Modulor and the 5 points (Le Corbusier)
    - the Barcelona Pavilion (Mies: "less is more")
    - Bauhaus (Gropius, 1919, Weimar)
    - the Sydney Opera House (Utzon)
    - the Louvre Pyramid (Pei)
    - Guggenheim Bilbao (Gehry, who died Dec 2025)

    - Heydar Aliyev Center (Hadid: the first woman Pritzker laureate, 2004)
    - the Gherkin (Foster)
    - the Pompidou (Piano + Rogers)
    - the Shard (Piano)
    - the Church of the Light (Ando)
    - paper shelters (Shigeru Ban)
    - the geodesic dome (Fuller)
    - the Burj Khalifa (SOM, 828 m)
    - the Bird's Nest (Herzog & de Meuron)
    - the Sagrada Família (Gaudí)
- **Recent Pritzker winners:** 2022 Kéré · 2023 Chipperfield · 2024 Yamamoto · 2025 Liu Jiakun · 2026 Smiljan Radić [reported].
- **Indian World Heritage Sites:**
    - 44th: the Maratha Military Landscapes (2025)
    - reported 45th: Sarnath (2026)
    - the Moidams (2024)
    - Hoysala and Santiniketan (2023)
    - Dholavira and Ramappa (2021)
    - Ahmedabad (the first WH City, 2017), Jaipur (2019)
    - Khangchendzonga (the only mixed site)

9. DESIGNERS & MOVEMENTS (rapid list)

- **Graphic and type:**
    - Paul Rand (IBM)
    - Milton Glaser (I ❤ NY)
    - Saul Bass (film titles)
    - Vignelli (the NYC subway map, 1972)
    - Harry Beck (the Tube map)
    - Edward Johnston (the Underground typeface, 1916)
    - Otl Aicher (the Munich 1972 pictograms)
    - Müller-Brockmann (grids)
    - Miedinger (Helvetica, 1957)
    - Renner (Futura, 1927)
    - Gill (Gill Sans)
    - Morison (Times New Roman, 1932)
    - Frutiger (Univers, Frutiger)
    - Carter (Verdana, Georgia)
    - Tschichold
    - Davidson (the Nike swoosh)
    - Janoff (the Apple logo)
    - Leader (the FedEx logo)
    - Kare (the Mac icons)
    - D. Udaya Kumar (the ₹ symbol)
    - Satyajit Ray (the Ray Roman typeface)
- **Product and furniture:**
    - Dieter Rams (Braun)
    - Jony Ive (Apple)
    - Raymond Loewy (MAYA)
    - Charles & Ray Eames (the Lounge Chair; *Powers of Ten*; *The India Report*, 1958)
    - Breuer (the Wassily chair)

    - Rietveld (the Red and Blue Chair)
    - Thonet (No. 14)
    - Jacobsen (the Egg chair)
    - Saarinen (the Tulip chair)
    - Panton
    - Starck (Juicy Salif)
    - Sottsass (Valentine; Memphis)
    - Ekuan (the Kikkoman bottle)
    - Yanagi (the Butterfly stool)
    - Noguchi (Akari)
    - Porsche (the Beetle)
    - Giacosa (the Fiat 500)
    - Issigonis (the Mini)
    - D'Ascanio (the Vespa)
- **Fashion:** Chanel (the LBD), Dior (the New Look, 1947), YSL (Le Smoking), Miyake (Pleats Please), Kawakubo (CDG), Westwood (punk), Armani (who died in 2025), Ritu Kumar, Sabyasachi, Rahul Mishra, Bhanu Athaiya (the first Indian Oscar, 1983).
- **Movements:**
    - Arts & Crafts (Morris)
    - Art Nouveau (Mucha, Gaudí)
    - De Stijl (Mondrian)
    - Bauhaus (1919–33)
    - Constructivism
    - Art Deco
    - Mid-century modern
    - Swiss style
    - Pop Art (Warhol)
    - Op Art (Riley)
    - Minimalism
    - Memphis (1981)
    - Postmodernism
    - Deconstructivism
    - flat design, then skeuomorphism, then Liquid Glass (2025)
- **Art movements:**
    - Impressionism (Monet)
    - Post-Impressionism (Van Gogh, Seurat's pointillism)
    - Expressionism (Munch)
    - Fauvism (Matisse)
    - Cubism (Picasso, Braque)
    - Futurism
    - Dada (Duchamp's *Fountain*)
    - Surrealism (Dalí, Magritte)
    - Abstract Expressionism (Pollock, Rothko)

10. LOGOS & PRODUCTS (rapid list)

- **FedEx:** an arrow between the E and the x. **Amazon:** an arrow from a to z (and a smile). **Toyota:** all the letters can be found in the emblem.
- **Baskin-Robbins:** a 31. **Toblerone:** a bear. **WWF:** closure. **Unilever:** 25 icons.
- **Audi:** 4 merged companies. **Mercedes:** land, sea, air. **BMW:** the Bavarian flag.
- **Olympics:** 5 rings. **Starbucks:** a siren. **Shell:** Loewy.

- **India:**
  - the Amul girl (1966)
  - the Air India Maharaja (1946) and "The Vista" (2023)
  - the national emblem (the Sarnath Lion Capital; *Satyameva Jayate* from the Mundaka Upanishad)
  - the ₹ symbol (2010)
  - Swachh Bharat (Gandhi's glasses)
  - Make in India (a lion made of cogs)
  - Incredible India (the ! as an i)
  - Gattu (R.K. Laxman)
- **Products:**
  - the Coca-Cola bottle (1915)
  - the Walkman (1979)
  - the iPhone (2007)
  - the Vespa (1946)
  - the Ambassador (Hindustan Motors)
  - the Tata Nano (2008)
  - the Jaipur Foot (1969)
  - the steel tiffin
  - the kulhad
  - LEGO (Denmark)
  - IKEA (Sweden)
  - MUJI (Japan)
- **Currency motifs:** ₹10 Konark · ₹20 Ellora · ₹50 Hampi · ₹100 Rani ki Vav · ₹200 Sanchi · ₹500 Red Fort.

11. KEY TERMS (one line each)

Term	Meaning
Affordance	An object's form suggests how to use it
Signifier	A cue that shows how to use something
Mapping	Controls correspond to their effects
Feedback	The system responds to an action
Poka-yoke	Mistake-proofing
Persona	A research-based user profile
Journey map	The user's steps with their pain points
Prototype	A test model
Iteration	Repeated refinement
Anthropometry	Body measurement
Biomimicry	Design inspired by nature
Upcycling	Turning waste into higher value
Greenwashing	False eco-claims
LCA	Life-cycle assessment
Skeuomorphism	UI that imitates real materials
Vector / raster	Paths vs pixels
Knolling	Arranging objects at right angles in a neat grid

Term	Meaning
Dieline	The flat net of a package
Isometric	A parallel projection with axes 120° apart
Chiaroscuro	Strong light–dark contrast
Vernacular	Local and traditional
Jaali	A perforated screen
Jharokha	An overhanging balcony window
Charbagh	A four-part garden
Shikhara / vimana / gopuram	Temple towers and gateway (Nagara tower / Dravida tower / southern gateway)

12. FLASHCARDS (160)

How to use them:

1. Cover the "Back" column and answer aloud.
2. Tick the cards you get right.
3. Re-test the unticked cards after 1, 3 and 7 days (spaced repetition).
4. You can paste them into Anki or onto index cards.

12.1 Design terminology & principles (1–25)

#	Front	Back
1	Shape vs form?	Shape is 2D; form is 3D (has volume)
2	Proportion vs scale?	Proportion compares parts within an object; scale compares size to a reference (often the human)
3	Three balance types?	Symmetrical, asymmetrical, radial
4	Progressive rhythm?	Repetition with a gradual change (e.g. growing sizes)
5	Hierarchy?	The order of importance shown through size, weight, colour and position
6	Negative space?	The empty space around and between subjects (the FedEx arrow)
7	Gestalt "closure"?	The mind completes incomplete shapes (the WWF panda)
8	Gestalt "proximity"?	Things that are near each other look grouped
9	Rule of thirds?	Place subjects on the third lines or their intersections
10	Lead room?	Space in front of a moving or looking subject
11	Stanford design thinking stages?	Empathise, Define, Ideate, Prototype, Test
12	Double Diamond?	Discover, Define, Develop, Deliver (Design Council, 2005)
13	Affordance example?	A cup handle suggests gripping; a flat door plate suggests pushing
14	Mapping example?	Stove knobs laid out like the burners
15	Poka-yoke example?	A SIM tray with a cut corner
16	Curb-cut effect?	Features designed for disabled people benefit everyone
17	Ergonomic percentile range?	The 5th to the 95th percentile
18	"Form follows function" coined by?	Louis Sullivan (1896)
19	"Less is more"?	Mies van der Rohe

#	Front	Back
20	Rams' motto?	"Less, but better" (*Weniger, aber besser*)
21	Loewy's MAYA?	Most Advanced Yet Acceptable
22	Norman's famous book?	*The Design of Everyday Things* (1988)
23	SCAMPER?	Substitute, Combine, Adapt, Modify, Put to other use, Eliminate, Reverse
24	Six Thinking Hats by?	Edward de Bono
25	Biomimicry example?	Velcro (burrs); the Eastgate Centre (termite mounds)

12.2 Colour (26–40)

#	Front	Back
26	RYB complement of blue?	Orange
27	RYB complement of yellow?	Violet
28	Tint, shade, tone?	Adding white, black and grey respectively
29	RGB: red + green light?	Yellow
30	CMY: cyan + magenta ink?	Blue
31	Analogous scheme?	Neighbours on the wheel
32	Triadic scheme?	Three hues 120° apart
33	Split-complementary of red?	Red + yellow-green + blue-green
34	60-30-10?	Dominant, secondary and accent colour shares
35	Atmospheric perspective?	Distance makes things paler, bluer and less detailed
36	Saturation?	The intensity or purity of a colour
37	ISO blue sign?	A mandatory action
38	ISO yellow sign?	A warning
39	Why not red text on green?	Low value contrast; fails for colour-blind viewers
40	Colour of mourning in many Indian communities?	White

12.3 Typography & communication (41–55)

#	Front	Back
41	Kerning?	The space between one specific letter pair
42	Tracking?	Uniform spacing across a word or line
43	Leading?	The space between lines
44	Legibility vs readability?	Telling characters apart vs reading blocks of text easily
45	Counter?	The enclosed space in a letter (o, a, e)
46	Helvetica designer and year?	Max Miedinger (with Eduard Hoffmann), 1957
47	Futura?	Paul Renner, 1927
48	Times New Roman?	Stanley Morison, 1932, for *The Times*
49	London Underground typeface?	Johnston (1916)
50	Frutiger was made for?	Charles de Gaulle Airport signage
51	Munich 1972 pictograms?	Otl Aicher
52	NYC subway map (1972)?	Massimo Vignelli
53	Tube map (1931/33)?	Harry Beck

#	Front	Back
54	Prohibition sign format?	A red circle with a diagonal slash
55	Max typefaces per layout?	2

12.4 Designers, artists & architects (56–90)

#	Front	Back
56	₹ symbol designer?	D. Udaya Kumar (2010)
57	NID founded?	1961, Ahmedabad (after the Eames' *India Report*, 1958)
58	NID founders (family)?	Gira and Gautam Sarabhai
59	NID became an INI by?	The NID Act, 2014
60	The Eames' admired Indian object?	The lota
61	Paul Rand logos?	IBM, ABC, UPS
62	I ❤ NY?	Milton Glaser (1977)
63	Nike swoosh?	Carolyn Davidson (1971)
64	Apple logo?	Rob Janoff (1977)
65	Mac icons?	Susan Kare
66	Braun designer?	Dieter Rams
67	iPod and iPhone designer?	Jony Ive
68	Wassily Chair?	Marcel Breuer (1925)
69	Red and Blue Chair?	Gerrit Rietveld
70	Barcelona Chair?	Mies van der Rohe
71	Egg Chair?	Arne Jacobsen
72	Tulip Chair?	Eero Saarinen
73	Juicy Salif?	Philippe Starck (Alessi, 1990)
74	Memphis Group founder?	Ettore Sottsass (1981)
75	Kikkoman bottle?	Kenji Ekuan (1961)
76	Fallingwater?	Frank Lloyd Wright
77	Villa Savoye?	Le Corbusier
78	Bauhaus founder?	Walter Gropius (1919, Weimar)
79	Sydney Opera House?	Jørn Utzon
80	Louvre Pyramid?	I.M. Pei (1989)
81	Guggenheim Bilbao?	Frank Gehry (1997)
82	First woman Pritzker laureate?	Zaha Hadid (2004)
83	First Indian Pritzker laureate?	B.V. Doshi (2018)
84	First African Pritzker laureate?	Diébédo Francis Kéré (2022)
85	Pritzker 2025?	Liu Jiakun (China)
86	Pritzker 2026?	Smiljan Radić Clarke (Chile) [reported]
87	IIM Ahmedabad architect?	Louis Kahn
88	Jawahar Kala Kendra architect?	Charles Correa
89	Low-cost brick architect in Kerala?	Laurie Baker
90	New Parliament architect?	Bimal Patel (HCP)

12.5 Movements & art (91–110)

#	Front	Back
91	Impressionism named after?	Monet's *Impression, Sunrise*
92	Pointillism?	Seurat
93	*The Starry Night*?	Van Gogh
94	*The Scream*?	Munch (Expressionism)
95	Cubism founders?	Picasso and Braque
96	*Guernica* subject?	The bombing of a Spanish town (1937)
97	Melting clocks?	Dalí, *The Persistence of Memory*
98	"Ceci n'est pas une pipe"?	Magritte
99	*Fountain* (a urinal)?	Duchamp (Dada, 1917)
100	De Stijl palette?	Primary colours + black and white, a grid (Mondrian)
101	Pop Art leader?	Andy Warhol
102	Op Art?	Bridget Riley, Victor Vasarely
103	Drip painting?	Jackson Pollock
104	Fauvism?	Matisse; wild colour
105	Arts and Crafts leader?	William Morris
106	Art Nouveau posters?	Alphonse Mucha
107	Chiaroscuro?	Strong light–dark contrast (Caravaggio)
108	Linear perspective pioneer?	Brunelleschi
109	Vitruvian Man?	Leonardo da Vinci
110	Progressive Artists' Group?	Bombay, 1947: Souza, Raza, Husain

12.6 Indian crafts, textiles & art (111–140)

#	Front	Back
111	Warli state and features?	Maharashtra; white rice paste on mud; circle, triangle, square
112	Madhubani features?	Bihar; double outlines; no empty space
113	Gond features?	MP; dots and dashes fill forms; Jangarh Singh Shyam
114	Phad?	Rajasthan scroll of Pabuji, performed by Bhopas
115	Kalighat?	Kolkata; bold brush lines; satire
116	Srikalahasti vs Machilipatnam Kalamkari?	Freehand pen vs block print
117	"Bani Thani"?	Kishangarh, by Nihal Chand ("Indian Mona Lisa")
118	Patola?	Patan; double ikat
119	Ikat key fact?	The yarn is dyed before weaving
120	Kanjeevaram border technique?	Korvai
121	Paithani motifs?	Peacocks and parrots
122	Baluchari?	Narrative scenes on the pallu (WB)
123	Muga silk?	Assam; naturally golden
124	Eri silk?	"Ahimsa" silk
125	Chikankari?	Lucknow; white-on-white embroidery
126	Phulkari?	Punjab; "flower work"

#	Front	Back
127	Kantha?	Bengal; running stitch on layered old saris
128	Ajrakh colours?	Indigo and madder red
129	Leheriya vs Bandhani?	Waves vs dots
130	Dabu?	Mud-resist printing
131	Blue Pottery base?	Quartz (not clay), Jaipur
132	Bidriware?	Bidar; blackened zinc alloy with silver inlay
133	Dhokra?	Lost-wax metal casting
134	Channapatna?	Karnataka; lacquered wooden toys
135	Aranmula Kannadi?	Kerala; a metal-alloy mirror
136	Longpi pottery?	Manipur; black; made without a wheel
137	Rogan art?	Castor-oil paint; Nirona (Kutch)
138	First GI in India?	Darjeeling Tea (2004)
139	GI Registry location?	Chennai
140	Garba UNESCO ICH year?	2023

12.7 Indian architecture & heritage (141–155)

#	Front	Back
141	Nagara vs Dravida tower?	Shikhara vs vimana (+ gopuram gateways)
142	Kailasa Temple?	Ellora Cave 16; carved top-down from one rock; Rashtrakuta Krishna I
143	Brihadeeswara?	Thanjavur; Rajaraja Chola I, about 1010
144	Konark?	A sun-temple chariot with 24 wheels; Narasimhadeva I
145	First garden tomb?	Humayun's Tomb
146	Taj Mahal chief architect?	Ustad Ahmad Lahauri
147	Gol Gumbaz?	Bijapur; a whispering gallery
148	Hawa Mahal?	Jaipur, 1799; Lal Chand Ustad
149	Rani ki Vav?	Patan stepwell; on the ₹100 note
150	CSMT architect?	F.W. Stevens (1888)
151	India's first WH City?	Ahmedabad (2017)
152	India's 44th WHS?	The Maratha Military Landscapes (2025)
153	Moidams?	Ahom mound burials, Assam (WHS 2024)
154	Only mixed WHS in India?	Khangchendzonga NP
155	Chandigarh planner?	Le Corbusier (with Pierre Jeanneret)

12.8 Logos, symbols & general awareness (156–160)

#	Front	Back
156	National emblem source?	The Lion Capital of Ashoka, Sarnath
157	Flag ratio and chakra spokes?	2:3; 24 spokes
158	Amazon logo arrow?	From a to z (and a smile)
159	Air India 2023 logo name?	"The Vista"
160	World Design Capital 2026?	Frankfurt RheinMain

13. TOP 100 NID DAT MISTAKES STUDENTS MAKE (and the fix for each)

A. Time management (1–12)

1. Spending 40+ minutes on the first drawing. **Fix:** write a time budget per question before starting.
2. Polishing one drawing and leaving the last question blank. **Fix:** use the 70% rule.
3. Not reading the whole paper first. **Fix:** spend 5 minutes reading everything.
4. Getting stuck on a hard MCQ. **Fix:** move on after 90 seconds.
5. No buffer time. **Fix:** keep 15–20 minutes for finishing touches.
6. Starting to colour before all the questions are drafted. **Fix:** colour last.
7. Not wearing a watch (where allowed). **Fix:** practise with the same watch.
8. Not practising full 3-hour papers. **Fix:** at least 5 timed mocks.
9. Erasing too much. **Fix:** start with light construction lines, then commit.
10. Rewriting text answers. **Fix:** draft mentally, then write once.
11. Underestimating multi-part questions. **Fix:** count the sub-tasks and split the time.
12. Panicking after one bad question. **Fix:** it's one of 14+ tasks; reset and move on.

B. Reading instructions (13–22)

13. Drawing 3 items when 5 are asked for. **Fix:** circle the numbers.
14. Ignoring the viewpoint (e.g. "from a crow's view"). **Fix:** underline the viewpoint words.
15. Using extra colours despite a limit. **Fix:** put the allowed colours in a separate pile.
16. Exceeding or falling short of the word limit. **Fix:** count the lines (about 10–12 words per line).
17. Missing "label" or "explain" instructions. **Fix:** tick each instruction when done.
18. Answering a different question (e.g. drawing a product when a poster was asked for). **Fix:** restate the task in the margin.
19. Ignoring the specified user (elderly, child). **Fix:** draw the user in the solution.
20. Drawing outside the box or margins. **Fix:** plan the layout first.
21. Not numbering the frames. **Fix:** always number them.
22. Forgetting the title or caption when asked. **Fix:** add a checklist line at the end.

C. Drawing & proportion (23–40)

23. People taller than doors. **Fix:** use size references.
24. Floating people. **Fix:** anchor their feet on the ground plane; add cast shadows.
25. Several horizon lines in one scene. **Fix:** one HL, drawn first.
26. Tilted verticals in 1- and 2-point perspective. **Fix:** use a scale for the verticals.
27. VPs too close together. **Fix:** put the VPs far apart, even off the page.
28. Stiff, mannequin figures. **Fix:** start with the line of action and gesture.
29. Identical faces and poses for every person. **Fix:** vary age, posture and clothing.
30. Tiny drawings in huge boxes. **Fix:** fill about 70–80% of the box.
31. Over-detailing one corner. **Fix:** spread the detail across the focal hierarchy.
32. Shading with no consistent light source. **Fix:** draw a small sun or arrow first.
33. Messy, hairy lines. **Fix:** ghost the stroke, then draw it confidently.
34. Not showing materials. **Fix:** use texture conventions for each material.
35. Hands hidden behind backs to avoid drawing them. **Fix:** practise mitten hands.
36. Ignoring depth layers. **Fix:** foreground, middle and background.
37. Wrong object details (e.g. a bicycle chain on the left). **Fix:** do weekly memory-vs-reality checks.
38. Drawing from imagination only, never from observation. **Fix:** daily sketching from life.
39. Perspective on circles (ellipses) going wrong. **Fix:** practise ellipses in boxes.

40. Rendering photorealistically at the cost of the idea. **Fix:** idea first.

D. Composition & communication (41–55)

41. Centring everything. **Fix:** use the rule of thirds.
42. No focal point. **Fix:** create one with contrast, size or isolation.
43. Clutter. **Fix:** remove elements that don't serve the story.
44. Tangents. **Fix:** overlap the objects or separate them clearly.
45. Poor hierarchy in posters. **Fix:** headline > image > text > logo.
46. Too much text on a poster or sign. **Fix:** 3–6-word headlines; use icons.
47. Illegible handwriting. **Fix:** print in capitals for labels.
48. Labels with no arrows. **Fix:** use thin leader lines.
49. Unclear storyboards. **Fix:** vary the shots; keep the characters consistent.
50. Changing the character's look between frames. **Fix:** give them one distinctive accessory.
51. A story with no resolution. **Fix:** plan the beats: setup, problem, action, resolution.
52. Ignoring negative space. **Fix:** let the subject breathe.
53. Random colours. **Fix:** choose a scheme (analogous, complementary).
54. Low contrast in signs. **Fix:** dark on light or light on dark.
55. A logo that fails in black and white. **Fix:** test a mono version.

E. Creativity & originality (56–68)

56. The first cliché (a crying Earth, a melting globe). **Fix:** generate 5 ideas and pick the 3rd or 4th.
57. Copying a famous logo or poster from memory. **Fix:** start from the brief's keywords.
58. Being "creative" but irrelevant. **Fix:** check the idea against the brief.
59. Only one category of ideas (low flexibility). **Fix:** vary the categories (tool, toy, art, shelter).
60. Overusing technology (apps, drones, robots). **Fix:** think physical, low-cost and contextual first.
61. Unrealistic magic solutions. **Fix:** hint at the mechanism or material.
62. Fear of simple ideas. **Fix:** a simple idea done brilliantly beats a complex mess.
63. Not using the Indian context. **Fix:** add local details (chai, autos, rangoli).
64. Ignoring humour or emotion where appropriate. **Fix:** a touch of wit makes answers memorable.
65. Not personalising "product you love" answers. **Fix:** a real story plus specific details.
66. Repeating the same idea style in every answer. **Fix:** vary your approaches.
67. Ignoring viewpoint as a creative lever. **Fix:** try unusual angles.
68. Not sketching thumbnails. **Fix:** 3 thumbnails in 3 minutes.

F. Problem-solving & user focus (69–80)

69. Jumping to a solution without defining the problem. **Fix:** write a one-line problem statement.
70. Designing for yourself, not the user. **Fix:** draw the user.
71. Ignoring constraints (cost, electricity, water). **Fix:** list the constraints first.
72. Solving one stakeholder's problem only (e.g. in the neem-branch dispute). **Fix:** consider every party.
73. Destructive solutions (ban it, cut it down). **Fix:** look for win-win, sustainable fixes.
74. Ignoring safety, especially for children's products. **Fix:** add one safety note.
75. Ignoring maintenance and cleaning. **Fix:** think about the whole life cycle.
76. Ignoring accessibility. **Fix:** consider low vision, mobility and literacy.
77. No explanation of how the solution works. **Fix:** labels plus a 2-line note.
78. No in-use scene. **Fix:** a small inset showing the user using it.
79. Over-complex systems for a small problem. **Fix:** proportionate solutions.
80. Not considering the systemic fix (e.g. repairing the pothole, not just helping the person who fell). **Fix:** end with the system improvement.

G. Visual reasoning & objective questions (81–92)

81. Forgetting to reverse the letter order in mirror words. **Fix:** reverse, then flip each letter.
82. Confusing a mirror image with a rotation. **Fix:** a rotation never produces a mirror image.
83. Unfolding in the wrong order. **Fix:** unfold in reverse.
84. Reflecting across the wrong axis in folding. **Fix:** draw the fold line.
85. Calling faces opposite when they are adjacent in nets. **Fix:** the one-apart rule.
86. Forgetting the downward triangles when counting. **Fix:** count systematically by size and orientation.
87. Mixing up the RGB and RYB wheels. **Fix:** check whether the question is about paint or light.
88. Overthinking simple GK. **Fix:** go with the first well-founded answer.
89. Leaving MCQs blank (there is no negative marking). **Fix:** guess intelligently.
90. Unit errors (cm vs m). **Fix:** convert first.
91. Accepting a syllogism conclusion that is only possible. **Fix:** it must be true in every diagram.
92. Careless option marking. **Fix:** re-check the question number against the answer sheet.

H. General awareness & preparation habits (93–100)

93. Memorising GK without visuals. **Fix:** look at images of every craft and building you learn.
94. Ignoring Indian crafts and textiles. **Fix:** 3 crafts a day from Vol 2C.
95. Cramming current affairs. **Fix:** link each event to a design idea.
96. Not keeping an error log. **Fix:** Vol 13 §6, updated daily.
97. Practising only favourite topics. **Fix:** follow the plan's rotation.
98. Not getting feedback on drawings. **Fix:** a weekly review by a teacher or design student.
99. Skipping sleep before the exam. **Fix:** 7–8 hours; no all-nighters.
100. Forgetting the admit card, ID or kit. **Fix:** pack the night before using the checklist.

NEXT VOLUME: Volume 13 — Study Plans, Daily Routines, Revision System & Error Log (`13-VOL13-STUDY-PLANS.md`)